

Confidence Intervals

We are interested in X which has pdf $f(x; \theta)$. The estimate of θ is by a statistics $T = T(X_1, \dots, X_n)$. It is unlikely $T = \theta$. When T is continuous, $P(T = \theta) = 0$. How much did T miss θ ?

- Suppose X is a random variable with unknown mean θ . Let $X_1, \dots, X_n$ be a random sample from the distribution of X and let $\bar{X}$ be the sample mean. Since $E(\bar{X}) = \theta$, $\bar{X}$ is a point estimator of θ . But how far is $\bar{X}$ to θ ? We will study the general case later. Now for a specific case where $X \sim N(\theta, \sigma^2)$. So, $\bar{X} \sim N(\theta, \sigma^2/n)$. $(\bar{X} - \theta)/(\sigma/\sqrt{n})$ has a standard normal distribution.

So,

$$\begin{aligned} 0.954 &= P(-2 < \frac{\bar{X} - \theta}{\sigma/\sqrt{n}} < 2) \\ &= P(\bar{X} - 2\frac{\sigma}{\sqrt{n}} < \theta < \bar{X} + 2\frac{\sigma}{\sqrt{n}}) \end{aligned}$$

It says that before the sample is drawn the probability that the random interval $(\bar{X} - 2\frac{\sigma}{\sqrt{n}}, \bar{X} + 2\frac{\sigma}{\sqrt{n}})$ traps θ is 0.954. After the sample is drawn the realized interval has either traps θ or it has not. But because of the high probability of success, namely 0.954, before the sample is drawn, we call this interval a 95.4% **confidence interval** for θ .

- Suppose $X_1, \dots, X_n$ are form $N(\mu, \sigma^2)$. Actually, $T = (\bar{X} - \mu)/(S/\sqrt{n})$ has a t distribution with $n - 1$ degrees of freedom. For $0 < \alpha < 1$, define $t_{\alpha/2, n-1}$ to be the upper $\alpha/2$ critical point of a t distribution with $n - 1$ degrees of freedom. Then

$$(\bar{x} - t_{\alpha/2, n-1}s/\sqrt{n}, \bar{x} + t_{\alpha/2, n-1}s/\sqrt{n})$$

is the $(1 - \alpha)100\%$ confidence interval for μ .

Let $\bar{X}$ be the mean of a random sample of size n from a distribution that is $N(\mu, 9)$. Find n such that $P(\bar{X} - 1 < \mu < \bar{X} + 1) = 0.90$, approximately.

Central Limit Theorem (CLT)

Let $X_1, \dots, X_n$ denote a random sample from a distribution with mean μ and finite variance σ^2 . Then

$$W_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

follows approximately $N(0, 1)$ when n is large.

As we further show in the next chapter, the result stays the same if we replace σ by the sample standard deviation S , that is

$$\frac{\bar{X} - \mu}{S/\sqrt{n}}$$

follows approximately $N(0, 1)$ when n is large.

- Suppose T is an estimator for θ_0 such that

$$\sqrt{n}(T - \theta_0) \xrightarrow{D} N(0, \sigma_T^2).$$

The parameter σ_T^2 is the asymptotic variance of $\sqrt{n}T$ and it is usually unknown. First, let us assume σ_T is known.

- let $Z = \sqrt{n}(T - \theta_0)/\sigma_T$ be the standardized random variable. Then Z is asymptotically $N(0, 1)$. Hence, $P(-1.96 < Z < 1.96) \doteq 0.95$. So,

$$\begin{aligned} 0.95 &\doteq P(-1.96 < Z < 1.96) \\ &= P(-1.96 < \frac{\sqrt{n}(T - \theta_0)}{\sigma_T} < 1.96) \\ &= P(T - 1.96 \frac{\sigma_T}{\sqrt{n}} < \theta_0 < T + 1.96 \frac{\sigma_T}{\sqrt{n}}). \end{aligned}$$

- The interval $(T - 1.96 \frac{\sigma_T}{\sqrt{n}}, T + 1.96 \frac{\sigma_T}{\sqrt{n}})$ is a function of the random variable T , we call it **random interval**. The probability that the random interval contains θ_0 is approximately 0.95. For any realized value of T , this interval is called 95% **confidence interval** for θ_0 . 0.95 = 95% is a **confidence coefficient**.
- Often we do not know σ_T . Suppose S_T is a consistent estimator of σ_T , by Slutsky's Theorem

$$\frac{\sqrt{n}(T - \theta_0)}{S_T} \xrightarrow{D} N(0, 1).$$

So, $(T - 1.96 \frac{S_T}{\sqrt{n}}, T + 1.96 \frac{S_T}{\sqrt{n}})$ would be a random interval with approximate probability 0.95 of covering θ_0 . We call $s_t/\sqrt{n}$ the **standard error** of T .

- Let $X_1, \dots, X_n$ be a random sample from the distribution of a random variable X with unknown mean μ and unknown variance σ^2 . Let $\bar{X}$ and S^2 denote the sample mean and sample variance, respectively. By CLT and Slutsky's theorem, $\sqrt{n}(\bar{X} - \mu)/S$ has an approximate $N(0, 1)$ distribution. Hence, an approximate 95% confidence interval for μ is

$$(\bar{x} - 1.96s/\sqrt{n}, \bar{x} + 1.96s/\sqrt{n}).$$

- There is nothing special about the confidence coefficient 0.95. Consider a general confidence coefficient $1 - \alpha$, where $0 < \alpha < 1$. Define $z_{\alpha/2}$ as the upper $\alpha/2$ quantile of a standard normal, i.e., $1 - \Phi(z_{\alpha/2}) = \alpha/2$. So the following interval being an approximate $(1 - \alpha)100\%$ confidence interval for θ_0 :

$$(t - z_{\alpha/2}s_T/\sqrt{n}, t + z_{\alpha/2}s_T/\sqrt{n})$$

- Think of $z_{\alpha/2}\sigma_T/\sqrt{n}$ as the error parts of the confidence interval.
 - $\alpha < \alpha^*$ implies $z_{\alpha/2} > z_{\alpha^*/2}$. Selection of higher values for confidence coefficients leads to larger error terms and hence longer confidence intervals.
 - Choosing larger sample size decrease the error part and hence to shorter confidence intervals.
 - Usually σ_T is some type of scale parameter of the underlying distribution. An increase in scale(noise level) generally results in larger error terms and hence longer confidence interval.

- Let X be a Bernoulli random variable with probability of success p . Suppose $X_1, \dots, X_n$ is a random sample from the distribution of X . Let $\hat{p} = \bar{X}$. By CLT, $\hat{P}$ is approximately $N(p, p(1-p)/n)$. Hence, an approximate $(1 - \alpha)100\%$ is

$$(\hat{p} - z_{\alpha/2} \sqrt{\hat{p}(1 - \hat{p})/n}, \hat{p} + z_{\alpha/2} \sqrt{\hat{p}(1 - \hat{p})/n}),$$

where $\sqrt{\hat{p}(1 - \hat{p})/n}$ is called the standard error of $\hat{p}$.

References

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- [4] *John E. Freund's Mathematical Statistics with Applications* by Irwin Miller Marylees Miller